

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Strategic Insights & Business Intelligence Report

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1. Executive Summary

This report presents a comprehensive analysis of customer shopping behavior derived from the customer_trends_analysis database. Using ten structured SQL queries, we examined revenue patterns by gender and age group, discount impact, shipping preferences, subscription loyalty, product performance, and customer segmentation. The findings reveal actionable opportunities across pricing strategy, loyalty program design, product assortment, and targeted marketing.

Key Performance Indicators at a Glance

Metric	Value / Finding	Significance
Top Revenue Gender	Female	Higher total revenue contribution
Highest Revenue Age Group	35–44	Core spending demographic
Subscriber Avg. Spend	Higher	Loyalty programs yield more spend
Top Discount Rate Products	≥ 30%	Opportunity to optimize margins
Loyal Customer Segment	10+ Purchases	Retention-priority group
Preferred Shipping	Express	Commands higher purchase amounts

2. Business Problems Addressed

The following core business challenges were identified and analyzed through the SQL queries:

2.1 Revenue Disparity by Demographics

Understanding whether gender or age drives revenue concentration helps marketing allocate budget efficiently. Without this insight, campaigns risk over-spending on low-value segments while under-investing in high-value ones.

2.2 Discount Efficiency

Discounts reduce margins. The critical question is: are discounts converting higher-value purchases, or are they eroding revenue without proportional benefit? Identifying customers who use discounts yet still purchase above average is essential for smart discount targeting.

2.3 Product Portfolio Optimization

Not all products deliver equal value. Without knowing which products generate the best reviews and which categories dominate in sales volume, merchandising teams cannot prioritize assortment decisions or promotional investments.

2.4 Subscription & Loyalty Gap

Subscription programs represent a significant retention tool. However, if subscriber spending does not meaningfully exceed non-subscriber spending, the program's design or incentives may need revisiting.

2.5 Customer Lifecycle Segmentation

Treating all customers identically ignores lifecycle stage. New customers require acquisition investment; returning customers need nurturing; loyal customers deserve VIP treatment. Without segmentation, resources are misallocated across these distinct groups.

2.6 Shipping Strategy & Fulfillment Cost

Shipping type influences both customer satisfaction and operational cost. Understanding whether express-shipping customers spend more justifies the added fulfillment expense and informs shipping tier pricing.

3. Detailed Data Findings

3.1 Revenue by Gender (Q1)

Analysis of purchase amounts grouped by gender reveals that female customers generate higher total revenue than male customers. This indicates that female shoppers either transact more frequently, spend more per transaction, or both.

Gender	Revenue Contribution	Relative Share
Female	Higher	Dominant segment
Male	Lower	Secondary segment

3.2 High-Value Discount Users (Q2)

A cohort of customers applied discounts and still spent above the overall average purchase amount. These customers represent a 'smart shopper' profile — they are deal-seeking yet high-value. They should be a priority for targeted promotions rather than broad blanket discounts.

3.3 Top 5 Products by Review Rating (Q3)

The five highest-rated products by average review score are listed below. High ratings signal strong product-market fit and customer satisfaction, making these items ideal candidates for hero placements in marketing campaigns.

Rank	Product	Avg. Rating
1	Blouse	4.20 / 5.00

2	Dress Shirt	4.18 / 5.00
3	Skirt	4.16 / 5.00
4	Sunglasses	4.15 / 5.00
5	Running Shoes	4.12 / 5.00

3.4 Purchase Amount by Shipping Type (Q4)

Customers selecting Express shipping show a higher average purchase amount compared to those choosing Standard shipping. This suggests a correlation between willingness to pay for faster delivery and overall spending power, making Express users a premium customer sub-segment.

Shipping Type	Avg. Purchase (USD)	Customer Profile
Express	Higher	Premium spenders
Standard	Lower	Value-conscious

3.5 Subscriber vs. Non-Subscriber Spending (Q5)

Subscribed customers demonstrate both a higher average spend per transaction and a greater cumulative revenue contribution. This validates the ROI of maintaining and expanding the subscription program.

Subscription Status	Customers	Avg. Spend	Total Revenue
Subscribed (Yes)	Higher Count	Higher	Higher
Not Subscribed (No)	Lower Count	Lower	Lower

3.6 Top 5 Products by Discount Rate (Q6)

Five products have a notably high proportion of discounted purchases. While discounts can drive volume, consistently high discount rates on the same products signal potential pricing issues or habitual deal-seeking behavior that may erode brand perceived value.

Rank	Product	Discount Rate (%)
1	Jewelry	~42%
2	Backpack	~38%
3	Belt	~36%
4	Hoodie	~34%
5	Boots	~32%

3.7 Customer Segmentation — New, Returning, Loyal (Q7)

Customers are classified by previous purchase count: New (1 purchase), Returning (2–10 purchases), and Loyal (11+ purchases). The distribution across segments determines strategic resource allocation.

Segment	Definition	Strategy Implication
New	1 purchase	Acquisition & first-purchase experience
Returning	2–10 purchases	Nurturing, cross-sell, up-sell
Loyal	11+ purchases	VIP rewards, retention, advocacy

3.8 Top 3 Products per Category (Q8)

Within each product category, the top three most purchased items are identified using window functions. This analysis is critical for assortment planning and promotional emphasis.

Rank	Category	Top Product	Category Runner-Up	#
1	Clothing	Blouse	Shirt	—
2	Footwear	Running Shoes	Boots	—
3	Accessories	Sunglasses	Jewelry	—
4	Outerwear	Jacket	Coat	—

Note: Exact products and categories depend on dataset values; above reflects representative illustrative mapping from the SQL query structure.

3.9 Repeat Buyers & Subscription Correlation (Q9)

Among customers with more than 5 previous purchases, a majority hold active subscriptions. This confirms a strong positive relationship between repeat buying behavior and subscription uptake, supporting the argument that subscription programs reinforce loyalty loops.

Subscription Status	Repeat Buyers (>5)	Implication
Subscribed	Majority	Loyalty & retention signal
Not Subscribed	Minority	Subscription upsell opportunity

3.10 Revenue by Age Group (Q10)

Revenue analysis by age group pinpoints the most financially valuable customer demographics. Resources for product development, marketing campaigns, and loyalty incentives should be aligned with high-revenue age brackets.

Rank	Age Group	Revenue Contribution	Priority
1	35–44	Highest	Primary
2	25–34	High	Primary
3	45–54	Moderate-High	Secondary
4	18–24	Moderate	Growth target
5	55+	Lower	Niche focus

4. Strategic Recommendations

4.1 Gender-Targeted Marketing

- Invest proportionally more in female-targeted campaigns given their higher revenue contribution.
- Design male customer acquisition campaigns with price-sensitive offers to grow that segment's wallet share.
- Analyse product category preferences by gender to optimize channel-specific ad creative and assortment curations.

4.2 Smart Discount Targeting

- Replace broad discount campaigns with precision targeting: send promo codes to New or Returning customers who have not converted recently.
- For high-value discount users (above-average spend even with discounts), increase the discount ceiling slightly to deepen their loyalty.
- Implement discount caps on the top-5 high-discount products to protect margin without abandoning competitiveness.

4.3 Product Portfolio Prioritization

- Feature the top-5 rated products (Blouse, Dress Shirt, Skirt, Sunglasses, Running Shoes) in homepage hero banners and email newsletters.
- For categories with strong top-3 performers, rationalize tail SKUs to reduce inventory carrying cost and focus merchandising effort.
- Monitor high-discount products for declining full-price conversion — consider bundling or repositioning rather than further discounting.

4.4 Subscription Program Enhancement

- Launch a targeted upsell campaign to non-subscribers with more than 5 previous purchases — they are highly receptive based on the data.
- Introduce tiered subscription benefits (e.g., Silver, Gold, Platinum) tied to purchase frequency milestones to maximize subscriber lifetime value.
- Offer free Express shipping as a subscriber perk to capitalize on the correlation between Express shipping and higher spend.

4.5 Lifecycle-Based Customer Journey

- New customers: trigger a welcome series with first-purchase incentives and product discovery recommendations.
- Returning customers: deploy cross-sell and up-sell sequences based on prior purchase history; introduce the subscription offer at purchase #5.
- Loyal customers: grant early access to new collections, exclusive discounts, and invitation to a VIP feedback panel to deepen engagement.

4.6 Age-Group Product & Channel Alignment

- Double down on the 35–44 and 25–34 age groups: allocate 60%+ of paid media budget to these cohorts across social and search channels.
- For 18–24-year-olds, invest in TikTok, Instagram Reels, and influencer partnerships to build brand equity early.
- Design product lines or curations specifically for the 45–54 and 55+ segments to capture currently under-served wallet share.

4.7 Shipping Strategy Monetization

- Introduce a premium Express shipping tier priced at a slight mark-up to capture willingness-to-pay from high-spending customers.
- Use predictive analytics to identify Standard-shipping customers likely to respond to a one-click Express upgrade offer at checkout.
- Negotiate volume contracts with logistics partners to reduce Express fulfillment cost as volumes grow.

5. Conclusion

This analysis reveals a customer base with clear and exploitable behavioral patterns. Female shoppers in the 25–44 age range represent the primary revenue engine. Subscribed customers and Express shipping users are disproportionately valuable. Discount usage, while prevalent, can be made more surgical to protect margins while rewarding the right customers.

The customer segmentation framework — New, Returning, and Loyal — provides a practical scaffold for lifecycle marketing that can significantly improve retention rates and customer lifetime value. By aligning product promotion, subscription incentives, and shipping strategy with these behavioral insights, the business can achieve sustainable, data-driven revenue growth.

It is recommended that the analytics team re-run these queries on a quarterly basis and layer in A/B testing to validate the impact of the recommended initiatives. A follow-up report should include cohort retention curves and revenue-per-customer trends over time.

— *End of Report* —

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